

GRASKOP, South Africa

WMO: 682870

Lat: 24.935S

Lon: 30.838E

Elev: 4711

StdP: 12.36

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.7	40.7	16.6	15.1	57.6	21.8	19.5	57.8	15.4	64.7	13.1	58.5	3.3	40	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.4	82.7	62.7	80.2	63.3	78.1	63.5	69.9	76.1	68.6	74.8	67.6	73.6	8.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	123.6	72.9	66.8	117.9	71.7	65.6	112.9	70.6	37.2	76.3	36.0	74.8	35.0	73.6	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	11.4	9.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.9	65.6	66.2	64.6	61.4	58.2	54.9	52.8	57.0	60.9	61.1	63.4	65.2
	DBStd	6.68	3.58	3.66	4.14	4.42	4.59	4.39	5.05	6.08	6.87	6.55	6.16	5.19
	HDD50	53	0	0	0	0	3	6	27	10	4	2	1	0
	HDD65	1875	36	27	56	123	215	303	377	257	159	158	101	63
	CDD50	4036	485	454	453	342	256	153	115	226	330	347	403	472
	CDD65	384	56	61	44	15	3	0	1	8	35	38	53	70
	CDH74	1839	186	158	143	59	20	4	2	63	276	302	328	298
	CDH80	257	18	11	13	2	0	0	0	3	49	58	58	45
(14) Wind	WSAvg	4.8	4.5	4.5	4.7	4.2	4.6	4.6	4.7	5.5	5.1	5.0	5.3	4.7
(15) Precipitation	PrecAvg	47.4	8.2	7.2	6.1	2.7	0.9	0.5	0.6	0.7	1.3	3.0	5.8	8.0
	PrecMax	89.5	15.8	28.1	13.7	6.8	4.3	2.6	1.9	4.0	5.8	5.6	14.8	16.9
	PrecMin	20.5	2.3	0.4	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.5	1.8	2.6
	PrecStd	14.4	3.8	6.5	3.5	1.6	1.0	0.7	0.5	0.8	1.3	1.4	2.7	3.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.3	81.3	81.6	79.2	76.5	74.0	72.5	79.7	85.1	85.4	85.0	84.6
		MCWB	67.2	68.3	65.8	62.4	58.0	53.5	50.0	55.2	56.3	61.5	63.9	67.7
	2%	DB	78.2	78.1	77.7	75.5	73.0	70.1	68.4	75.7	80.7	80.8	80.7	80.2
		MCWB	67.5	67.1	65.4	61.7	56.9	52.4	49.5	54.1	56.3	62.0	64.1	67.3
	5%	DB	75.8	75.6	74.9	72.0	70.2	67.0	65.2	72.0	76.9	77.3	77.5	77.1
		MCWB	66.7	66.2	65.1	61.9	56.4	51.3	49.0	53.0	56.7	61.1	63.8	66.5
	10%	DB	73.2	73.4	72.3	69.0	67.3	64.0	61.8	68.3	73.0	73.3	74.3	74.3
		MCWB	66.0	65.6	64.6	61.2	55.7	50.6	48.7	52.3	56.8	60.2	63.1	65.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.3	70.7	70.6	68.0	63.0	57.4	55.5	60.1	64.2	67.7	69.2	71.5
		MCDB	76.1	76.9	75.8	73.4	69.5	67.3	62.3	71.0	73.1	77.9	77.8	78.5
	2%	WB	69.3	69.0	68.4	65.3	60.4	55.0	53.5	57.9	61.8	65.2	67.2	69.3
		MCDB	74.7	75.3	73.9	70.6	67.6	64.7	60.5	67.8	72.2	74.4	75.3	76.1
	5%	WB	68.1	67.7	66.7	63.7	58.9	53.5	51.9	56.2	60.1	63.3	65.7	68.0
		MCDB	73.1	73.8	72.4	68.8	66.4	63.4	59.8	65.4	70.1	72.3	73.5	74.4
	10%	WB	66.9	66.4	65.2	62.3	57.4	52.3	50.3	54.6	58.4	61.5	64.2	66.6
		MCDB	71.7	71.9	70.5	67.1	64.0	60.9	58.0	63.5	68.2	70.1	71.9	73.2
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	13.4	13.6	13.9	18.2	18.9	17.2	18.7	18.9	16.7	15.7	13.6
		MCDBR	18.4	17.6	18.9	19.1	21.4	22.2	21.8	24.3	24.9	24.0	22.8	19.5
	5% WB	MCWBR	9.6	8.9	9.2	9.9	10.1	10.4	9.9	9.7	9.8	10.0	10.2	9.7
		MCWBR	14.8	15.4	16.2	15.5	18.6	19.4	18.2	19.5	20.2	19.7	18.9	17.4
(40) Clear-Sky Solar Irradiance	taub		0.366	0.360	0.355	0.340	0.309	0.293	0.294	0.346	0.395	0.408	0.379	0.379
		taud	2.429	2.449	2.452	2.452	2.500	2.524	2.508	2.350	2.240	2.250	2.369	2.384
	Ebn at Noon		310	307	299	290	286	285	289	283	282	289	304	306
		Edh at Noon	39	38	36	33	29	26	28	36	43	45	42	41
(44) All-Sky Solar Radiation	RadAvg		1956	1914	1703	1455	1350	1218	1295	1502	1702	1753	1855	1929
	RadStd		143	161	134	103	61	73	43	87	104	117	158	145

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page